



In silico identification and expression analysis of MscS like gene family in rice

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ABSTRACT

Mechanosensitive channels are membrane proteins that open and shut in response to mechanical forces produced by osmotic pressure, sound, touch and gravity. These channels are involved in multiple physiological functions including hypoosmotic pressure, pain, hearing, blood pressure and cell volume regulation. In plants, these channels play a major role in proprioception, gravity sensing and maintenance of plastid shape and size. In the present study, we identified the mechanosensitive channel of small conductance like (MscS) homologue gene family in rice and analyzed their structure, phylogenetic relationship, localization and expression pattern. Five MscS like genes of rice (OsMSL) were found to be distributed on four chromosomes and clustered into two major groups. Subcellular localization predictions of the OsMSL family revealed their localization to plasma membrane, plastid envelope and mitochondria. The predicted gene structure, bonafide conserved signature motif, domain and the presence of transmembrane regions in each OsMSL strongly supported their identity as members of MscS-like gene family. Furthermore, *in silico* expression analysis of OsMSL genes revealed differential regulation patterns in tissue specific and abiotic stress libraries. These findings indicate that the *in silico* approach used here successfully identified in a genome-wide context MscS like gene family in rice, and further suggest the functional importance of MscS-like genes in rice.

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1. Introduction

Ion channels are multipass transmembrane proteins that form small aqueous pores across a membrane and allow ions to flow down their electrochemical gradient. They are ubiquitously present in all organisms and exhibit a significant role as sensor and effector molecules. Ion channels can be broadly classified into voltage gated, ligand gated and mechanical gated ion channels. A voltage gated ion channel is regulated by voltage difference across a membrane and a ligand gated ion channel responds to a ligand–receptor interaction and mechanosensitive (MS) ion channel sense and responds to mechanical stimuli. MS channels recognize a wide range of mechanical stimuli, and play important role in many physiological functions including osmotic pressure, touch and pain sensation, hearing, blood pressure, cell volume regulation, proprioception and gravity sense (Martinac, 1993; Sackin, 1995; Sachs and Morris, 1998; Hamill and Martinac, 2001). Prokaryotic MS channels have been extensively studied in *Escherichia coli* which harbors several

MS channels (Martinac et al., 1987; Berrier et al., 1989; Delcour et al., 1989; Booth et al., 2007; Edwards et al., 2012; Reuter et al., 2014). Mechanosensitive channels have been classified based on their ion conductance as mechanosensitive channels of large conductance (MscL), mechanosensitive channels of small conductance (MscS), mechanosensitive channels of mini conductance (MscM) and mechanosensitive channels of K⁺-selective (MscK) (Sukharev et al., 1994; Berrier et al., 1996). MS channels in bacteria also serve as emergency safety valves during hypoosmotic shock, which provide a graded series of response to osmotic stress in different environmental and developmental conditions (Martinac et al., 1987; Levina et al., 1999; Booth et al., 2005; Booth and Blount, 2012).

Membrane topology of *E. coli* MscL channel predicted the formation of transmembrane helices TM1 and TM2 (Sukharev et al., 1994, 1999; Blount et al., 1996a,b). MscL channel is a transmembrane protein which exposed N-terminal towards cytoplasmic interface of plasma membrane and C-terminal faces the cytoplasm (Steinbacher et al., 2007; Iscla et al., 2008). Similarly, the structure of *E. coli* MscS channel revealed that this protein is a homoheptamer with each subunit containing a membrane domain and three transmembrane regions of TM1, TM2 and TM3 helices. TM3 helix is highly conserved throughout domains as compared with TM1 and TM2 of MscL and functionally equivalent to TM1 of MscL channels (Bass et al., 2002; Balleza and Gómez-Lagunas, 2009).

Abbreviations: OsMSL, *Oryza sativa* MscS like; MscS, mechanosensitive channel small conductance; MscL, mechanosensitive channel large conductance; AtMSL, *Arabidopsis thaliana* MscS like.

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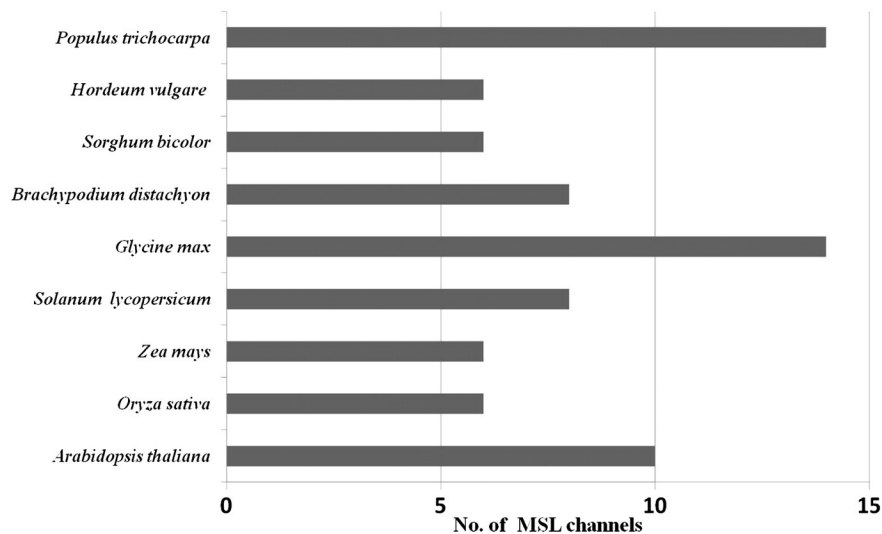


Fig. 1. MscS like (MSL) genes identified in different plant species. The number at the x-axis indicates the number of identified MSL genes.

The TM3 is rich in hydrophobic residues and line the channel pore (Miller et al., 2003). The C-terminal domain of MscS contains three subdomains, alpha-beta and beta barrel (Balleza and Gómez-Lagunas, 2009). The MscS channel family homologues are ubiquitously distributed in bacteria, archaea, protists, fungi, single-celled algae *Chlamydomonas* and in the plant *Arabidopsis thaliana* (Martinac et al., 1987; Gustin et al., 1988; Kloda and Martinac, 2001, 2002; Martinac and Kloda, 2003; Pivetti et al., 2003; Nakayama et al., 2007). MscS channel homologues are restricted to bacteria, archaea and fungi. MscS like channels are well characterized in the model plant *A. thaliana*. *Arabidopsis* has ten MscS homologues, called MscS like (MSL) and further classified into two main classes, class I and class II (Pivetti et al., 2003; Haswell, 2007). Class I MSL shows higher sequence resemblance with bacterial MscS channels and localized in either plastid envelop or mitochondria (Haswell, 2007). On the other hand, class II MSL like are restricted to plant, fungi and slime mold lineage and localized in plasma membrane. Similarly, other plant species such as *Populus trichocarpa*, *Brassica campestris*, *Brachypodium sylvaticum*, *Beta vulgaris* and *Lycopersicon esculentum* have MscS homologues (Pivetti et al., 2003; Haswell, 2007).

Molecular and electrophysiological characterization of *Arabidopsis* MSL (MSL2, MSL3, MSL9 and MSL10) have been extensively studied. Studies of the subcellular localization of MSL9 and MSL10 revealed their presence in the plasma membrane of root cells (Haswell et al., 2008). AtMSL10 channel has moderate selectivity for anions over cation with conductance of about 100 pS and reversibly inhibited by Gadolinium (Gd^{3+}) (Maksaev and Haswell, 2012). Similarly, AtMSL2 and AtMSL3 are localized in plastids and involved in various physiological activities such as maintenance of plastid shape and size, and placement of the divisional ring (Haswell and Meyerowitz, 2006; Wilson et al., 2011; Wilson and Haswell, 2012; Velej and Haswell, 2012). The functional analysis of *Arabidopsis* MSL2 showed that the conserved motif (PN (X)₉ N) are critical for maintenance of normal plastid and leaf morphology (Jensen and Haswell, 2012).

In the present study, we identified the MSL gene family across a diverse group of species. The work also involves the identification of rice MSL gene family and analysis of their gene structure, conserved motifs and phylogenetic relationships. We identified five MscS-like homologues in rice, which can be further grouped into two classes. MscS-like genes OsMSL1 and OsMSL2 represented class I and OsMSL4, OsMSL5, and OsMSL6 clustered into class II. The subcellular localization

predicted that class I OsMSL was localized either in the plastid membrane or mitochondria, whereas class II OsMSL was localized in the plasma membrane. By taking the advantage of available expression data for the rice MSL genes, we also performed an analysis of tissue specific expression of the rice MSL genes, underlying its interesting role in plant development and under different abiotic stresses. All OsMSL preferably demonstrated significant expression in reproductive stages, but in abiotic stresses, no significant regulations of transcripts were found.

2. Material and methods

2.1. Sequence and database search for MSL

MscS like (MSL) protein sequences were obtained from the NCBI (www.ncbi.nlm.nih.gov/) protein database with MscS-like as the query search. The search for plant MSL gene was carried out using BLASTP and 50% identity was taken as the threshold. The obtained sequences were searched in various plant genome databases such as TAIR (<http://www.arabidopsis.org/>), NCBI (www.ncbi.nlm.nih.gov/), Gramene (<http://www.gramene.org/>), Rap-db (<http://rapdb.dna.affrc.go.jp/>), TIGR (<http://rice.plantbiology.msu.edu/>), Phytozome (<http://www.phytozome.net/>), PlantGDB (www.plantgdb.org/) and Uniprot (www.uniprot.org/). 79 MSL sequences from various plant species which included *A. thaliana*, *Oryza sativa*, *Zea mays*, *Solanum lycopersicum*, *Glycine max*, *Brachypodium distachyon*, *Sorghum bicolor*, *Hordeum vulgare* and *P. trichocarpa* were retrieved. All MSL sequences were aligned using CLUSTAL 2.0.3 (<http://www.ebi.ac.uk/clustalw/>) multiple sequence alignment and a Hidden Markov Model (HMM) profile was constructed. Non-redundant protein sequences were searched using HMMER (hmmer.janelia.org/) (Finn et al., 2011). HMMER and BLAST hits were compared and manually edited to avoid redundant sequences. Amino acid sequences were aligned using CLUSTAL 2.0.3 (<http://www.ebi.ac.uk/clustalw/>) multiple sequence alignment tool and a phylogenetic tree was constructed by neighbor joining tree using MEGA 6.0 software with default parameters (Tamura et al., 2013). All predicted MSL sequences were analyzed for their subcellular localization *in silico* by LOCTREE3 predictive system (<https://www.rostlab.org/services/loctree3>) (Goldberg et al., 2014) and TargetP 1.1 prediction server (<http://www.cbs.dtu.dk/services/TargetP/>) (Emanuelsson et al., 2000).

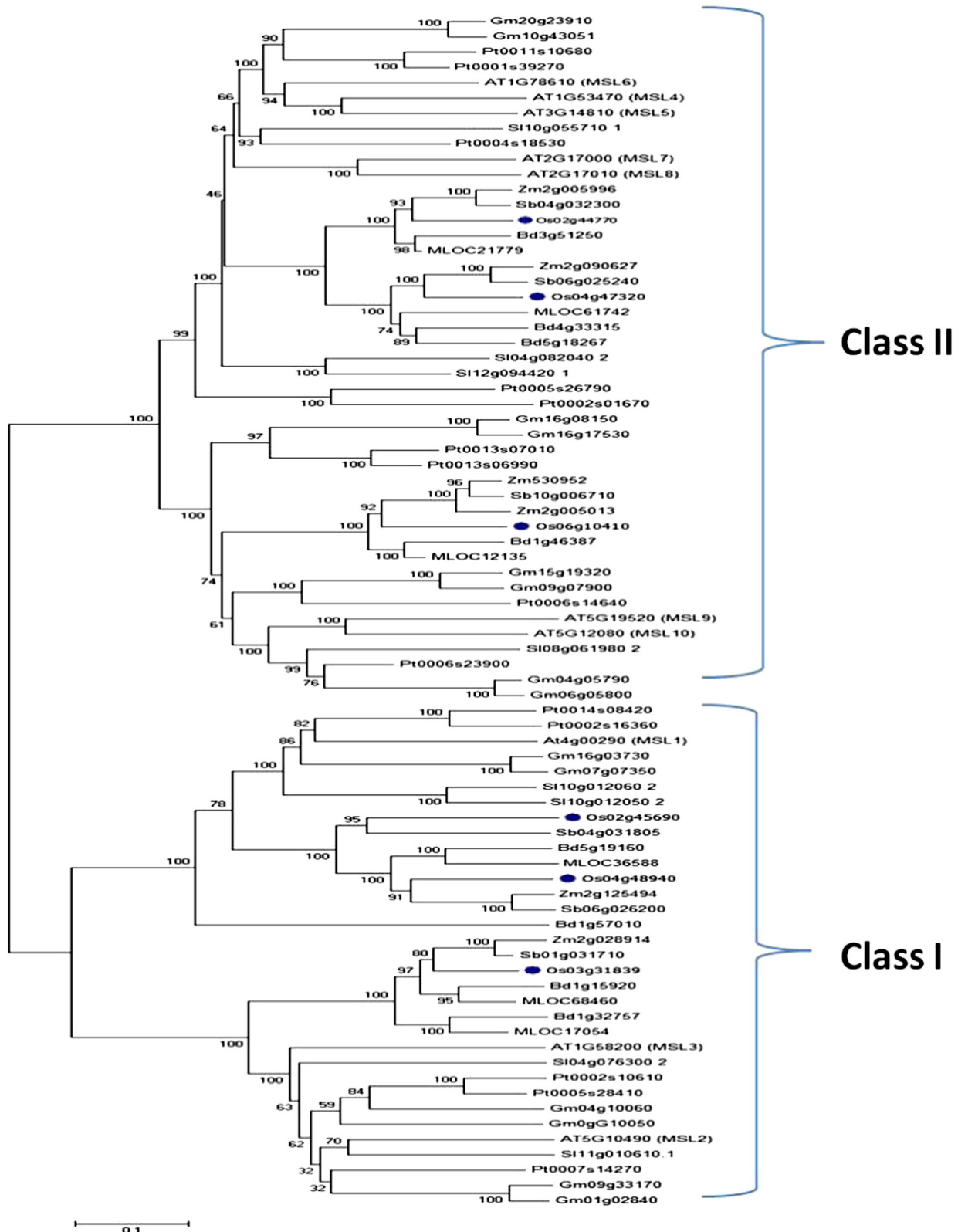


Fig. 2. Phylogenetic tree of the MscL like (MSL) channels among rice and other plant species, based on amino acid sequences. The tree was drawn according to results generated by MEGA 6.0 analysis using the neighbor-joining method with an amino acid and Poisson correction model. *Oryza sativa* MSL are marked by filled circle. Locus IDs of all MSL genes are given as: *Oryza sativa* (Os04g48940, Os02g45690, Os02g44770, Os06g10410, Os03g31839, Os04g47320), *A. thaliana* (At4g00290, At5g10490, At1g58200, At1g53470, At3g14810, At1g78610, At2g17000, At2g17010, At5g19520, At5g12080), *Zea mays* (Zm2g028914, Zm2g090627, Zm2g005013, Zm530952, Zm2g005996, Zm2g125494), *Brachypodium* (Bd5g19160, Bd3g51250, Bd1g15920, Bd1g32757, Bd4g33315, Bd5g18267, Bd1g46387, Bd1g57010), *Glycine max* (Gm16g03730, Gm04g10060, Gm20g23910, Gm10g43051, Gm15g19320, Gm09g07900, Gm04g05790, Gm06g05800, Gm16g08150, Gm16g17530, Gm09g33170, Gm01g02840, Gm0gG10050, Gm07g07350), *Solanum lycopersicum* (Sl10g012060, Sl10g012050, Sl04g076300.2, Sl11g010610.1), *Sorghum bicolor* (Sb04g031805, Sb01g031710, Sb10g006710, Sb04g032300, Sb06g025240), *Hordeum vulgare* (MLOC36588, MLOC61742, MLOC21779, MLOC68460, MLOC17054), *Populus* (Pt0014s08420, Pt0002s16360, Pt0002s10610, Pt0005s28410, Pt0007s14270, Pt0005s26790, Pt0013s07010, Pt0006s23900, Pt0013s06990, Pt0002s01670).

Table 1

Physicochemical characteristics of OsMSL genes. OsMSL—*Oryza sativa* MscS like, MSU—MSU Rice Genome Annotation Project, Chr. no.—Chromosome Number, CDS—coding sequences, AA—Amino acid, bp—Base pair, MW—Molecular weight, g/mol—Gram/mole, pI—Isoelectric point, mitochondria^P—TargetP 1.1 server localization prediction.

OsMSL	MSU ID	Chr. no./CDS coordinates	Nucleotide cds bp	AA	MW g/mol	pI	Subcellular localization
OsMSL1	Os04g48940	4/29187055–29193584	1575	524	56,567	8.3	Chloroplast membrane
OsMSL2	Os02g45690	2/27798048–27803943	1533	510	56,127	8.3	Mitochondria ^P or chloroplast membrane
OsMSL3	Os03g31839	3/18211372–18222645	5514	1838	210,326	6	Chloroplast membrane
OsMSL4	Os02g44770	2/27117991–27122440	2925	974	108,371	9.5	Plasma membrane
OsMSL5	Os04g47320	4/28091139–28095005	2805	935	103,874	9.2	Plasma membrane
OsMSL6	Os06g10410	6/5348679–5345377	2238	745	83072	10	Plasma membrane

2.2. Annotation of OsMSL on rice genome

Rice OsMSL sequences were analyzed for the presence of MS domain using InterProScan 4 (Zdobnov and Apweiler, 2001) (<http://www.ebi.ac.uk/Tools/pfa/iprscan/>) and SMART (Schultz et al., 2000) databases. Rice MscS like (OsMSL) genes were mapped on rice chromosome according to their genome coordinates.

2.3. OsMSL sequence analysis

Gene structure, exons and introns were obtained by comparing open reading frame (ORF) and genomic sequences. Structures were displayed using GSDS 2.0 (<http://gsds.cbi.pku.edu.cn/>) (Guo et al., 2007). The transmembrane regions with cartoon diagrams were predicted by Pspred MEMSAT-SVM (Buchan et al., 2013). The conserved motifs were identified using MEME suite (<http://meme.nbcr.net/meme/>) using default parameters (Timothy et al., 2009).

2.4. Gene expression data analysis

In silico expression profile of OsMSL genes were analyzed in tissue specific and abiotic stress libraries using RiceXPro (<http://ricexpro.dna.affrc.go.jp/>) (Sato et al., 2011, 2013) and Rice Oligonucleotide Array Database (ROAD) (www.ricearray.org/) databases (Cao et al., 2012) by retrieving the expression values from the array database. To gain insight into tissue specific expression profiles of OsMSL members in rice, transcript expressions were searched against RiceXpro databases using the MSU locus ID. Similarly we obtained abiotic stress expression profile data from the ROAD database. The heatmap was generated with the

software CIMMiner (<http://discover.nci.nih.gov/cimminer>) (Weinstein et al., 1994, 1997).

3. Result and discussion

3.1. Identification of MSL from plant species

We performed a genome-wide analysis of MSL genes in nine plants species that represent two major classes (monocot and dicot) and analyzed the distribution pattern of the MscS-like homologues. This study revealed the distribution of 6 to 8 MSL genes in the monocot group consisting of rice, maize, sorghum, barley and *Brachypodium*. The dicot group consisting of soybean and poplar has 14 MSL genes each, while tomato has only 8 MSL genes. Ten MSL genes were predicted in *Arabidopsis* (Haswell, 2007) (Fig. 1). The number of MSL gene distribution in dicots is nearly double than that of monocots, possibly due to the higher rate of genome duplication events in dicots. Four different major genome duplication events were postulated for *Arabidopsis* over 100 to 200 million years ago (Vision et al., 2000; Simillion et al., 2002). In plants, only MscS homologues are present, commonly known as MscS like (MSL) channels (Pivetti et al., 2003). However a group of Ca^{2+} -permeable mechanosensitive channels have been reported in the *A. thaliana* named as MCA1 and MCA2, are known to be involved in mechanical stress-induced Ca^{2+} influx (Nakagawa et al., 2007; Yamanaka et al., 2010; Shigematsu et al., 2014). Similarly, rice OsMCA1 is involved in a generation of reactive oxygen species (ROS) and hypoosmotic signaling in the rice (Kurusu et al., 2012a,b).

To analyze the phylogenetic relationship between MSL gene family members from various plant species, a phylogenetic tree was constructed using MEGA 6.0 software. Phylogenetic analysis demonstrated that plant MSL genes were clustered into 2 classes, namely class I and class II. Rice, *Arabidopsis*, sorghum and the barley genomes have 3 members of class I MSL, while *Brachypodium* and tomato genome has 4 members of class I MSL. Maize, poplar and soybean showed the presence of 2, 5 and 8 members of class I MSL respectively (Fig. 2). On the other hand, rice, *Arabidopsis*, sorghum and barley have 3 members each of class II MSL, while *Brachypodium*, maize and tomato showed the presence of 4 members of class II MSL. Soybean and poplar has 6 and 9 members of class II MSL respectively. Previous studies showed that MSL genes are categorized into two main classes, class I members exhibit resemblance with the prokaryotic MscS channels and class II form independent plant/fungi lineages (Pivetti et al., 2003; Haswell, 2007).

3.2. *In silico* analysis of MSL channels in rice

In our study, 6 MSL genes (designated as OsMSL1–6) were predicted in the rice genome according to their sequence similarity with *Arabidopsis* MSL. The average physicochemical parameters of OsMSL genes were included in the study such as protein length ranges from 510 to 1838 amino acids, molecular weight ranges from 56127 to 210325 g/mol and isoelectric point (pI) ranges from 6 to 10 (Table 1). Class I OsMSL genes (OsMSL1, OsMSL2 and OsMSL3) were localized either in chloroplast membrane or mitochondria and Class II OsMSL genes (OsMSL4, OsMSL5 and OsMSL6) were present in the plasma

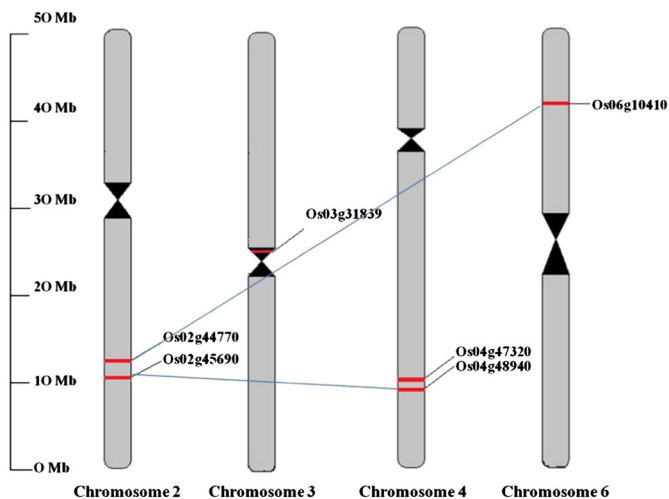


Fig. 3. Chromosomal location and duplication event schematic for OsMSL genes. Chromosome numbers are indicated at the bottom of each bar. The OsMSL genes represented by duplicated chromosomal segments between two chromosomes were connected by lines. The chromosomal positions of putative OsMSL were mapped according to the genome coordinates.

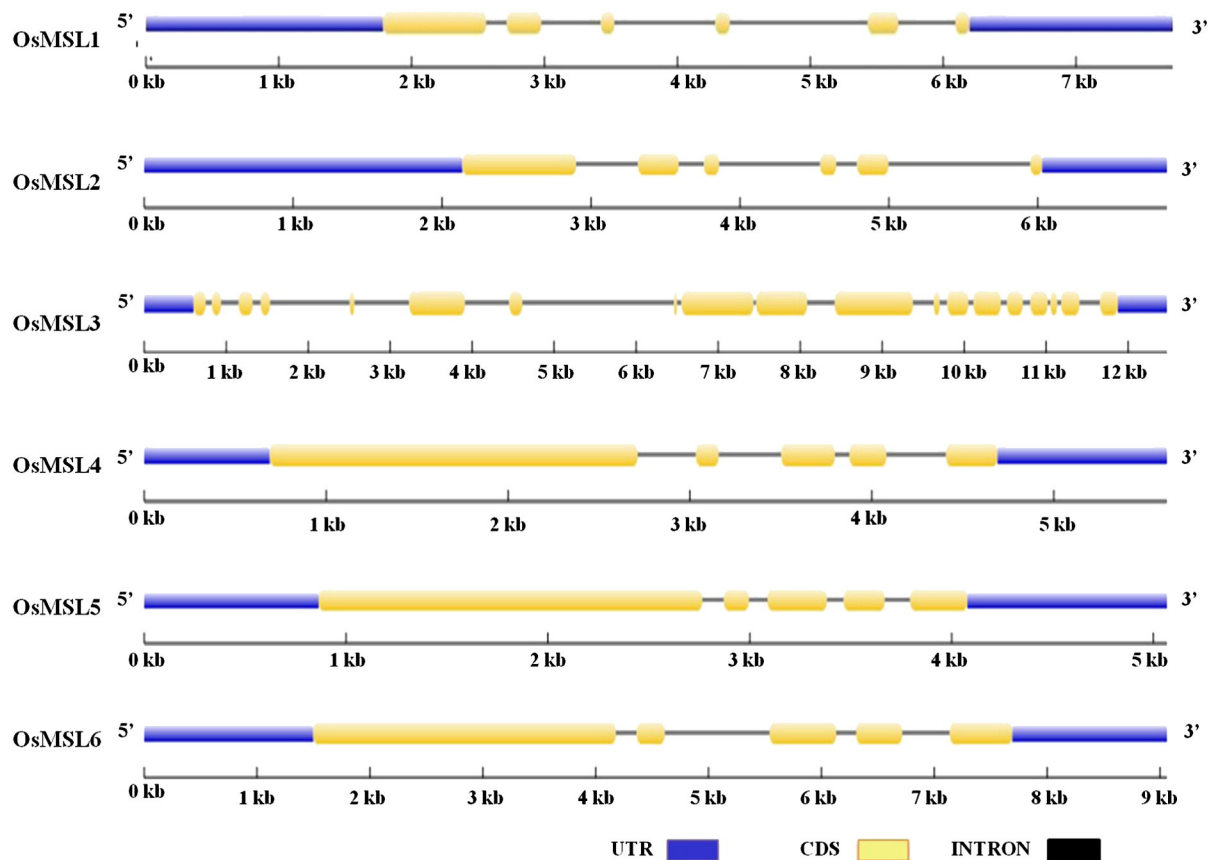


Fig. 4. Gene structure schematic diagram for OsMSL genes. Exons were demonstrated by filled yellow boxes and introns were demonstrated by black lines. Untranslated region (UTR) was displayed by filled blue boxes at both ends.

membrane (Table 1). Subcellular localization study in *A. thaliana* demonstrated that MSL2 and MSL3 were localized in chloroplast membrane (Haswell and Meyerowitz, 2006), whereas, AtMSL9 and AtMSL10 were found in the plasma membrane of root cells (Haswell et al., 2008). Class I MSL genes reported to be located either in plastid envelop or mitochondria (Haswell, 2007). Similarly in *Chlamydomonas reinhardtii* mechanosensitive channel (MSC1) is localized in a plastid envelope (Nakayama et al., 2007).

The physical location of the OsMSL genes could be assigned to the rice chromosomes using mapped coordinates of the rice genomic database (<http://rice.plantbiology.msu.edu/>) (Fig. 3). All rice OsMSL genes were localized to chromosomes 2, 3, 4 and 6. Both OsMSL2 and OsMSL4 genes were localized to chromosome 2. OsMSL1 and OsMSL5 were localized to chromosome 4. While OsMSL3 and OsMSL6 genes were localized to chromosomes 3 and 6, respectively, as shown in Fig. 3. Based on both phylogenetic relationship and sequence similarity, we have observed 2 pairs of OsMSL genes with high levels of sequence similarity. The protein sequences of OsMSL1 and OsMSL2 shared 70% sequence similarity, while OsMSL4 and OsMSL6 shared 45% sequence resemblance (Fig. 3). OsMSL1 and OsMSL2 genes are paralogues of each other, similarly OsMSL4 and OsMSL6 are paralogues of each other, which might have evolved by duplication events. OsMSL homologues in *Arabidopsis* were also identified based on sequence similarity and phylogenetic relationship. OsMSL1 and OsMSL2 are homologues of AtMSL1 and OsMSL3 is a homologue of AtMSL2 and AtMSL3. OsMSL4 is a homologue of AtMSL4, AtMSL5, AtMSL6, AtMSL7, and AtMSL8, whereas OsMSL6 is a homologue of AtMSL9 and AtMSL10 (Supplementary information Table S1).

We also determined the intron–exon boundaries for the 6 OsMSL genes to obtain more information on the genomic organization of rice OsMSL genes (Fig. 4). According to the annotation of the rice genome,

all OsMSL genes have introns in their structure and the number of exons varied from 5 to 19, and the highest numbers of exons were observed in OsMSL3.

The predicted topology of OsMSL genes using Pspired (MEMSAT-SVM) were illustrated in Fig. 5. OsMSL1 and OsMSL2 demonstrated the presence of five transmembrane (TM) regions, while OsMSL4, OsMSL5 and OsMSL6 exhibited the presence of six transmembrane regions. Surprisingly, OsMSL3 exhibits the presence of only one TM domain. However, OsMSL3 gene predicted topology is not consistent with inclusion in the OsMSL gene family, and is more likely to indicate a partial sequence, fusion event, or a transposon. In *E. coli* MscS channel topologies varied from 3 to 11 putative transmembrane segments (TMS) and adopting an N-terminal out and C-terminal in configuration (Miller et al., 2003; Pivetti et al., 2003). Again this was similar with *Arabidopsis* class I members, which showed the presence of five TM domains, while class II members exhibit the presence of six TM domains (Haswell, 2007).

3.3. Conserved motif and domain analysis

Amino acid alignment of MSL encoding genes from various organisms showed highly conserved regions. The MEME suite was used to analyze the conserved motifs in the OsMSL. These motifs could further provide deeper understanding that could help in gaining insights on the evolutionary relationships of the plant MSL family. Conserved motif (PF(X_{12–16})GXV(X_{20–21})PN(X₉)N) was identified in rice OsMSL at the C-terminal TM domain of class I MSL (OsMSL1, OsMSL2 and OsMSL3). Class II OsMSL (OsMSL4, OsMSL5 and OsMSL6) showed the presence of (F(X₃)P(X₃)GD(X_{10–14})V(X_{20–21})PN(X₇)IXNXXR) conserved motif at the C-terminal TM domain (Fig. 6). Eukaryotic MSL gene family was categorized into two main classes. Class I MSL aligned

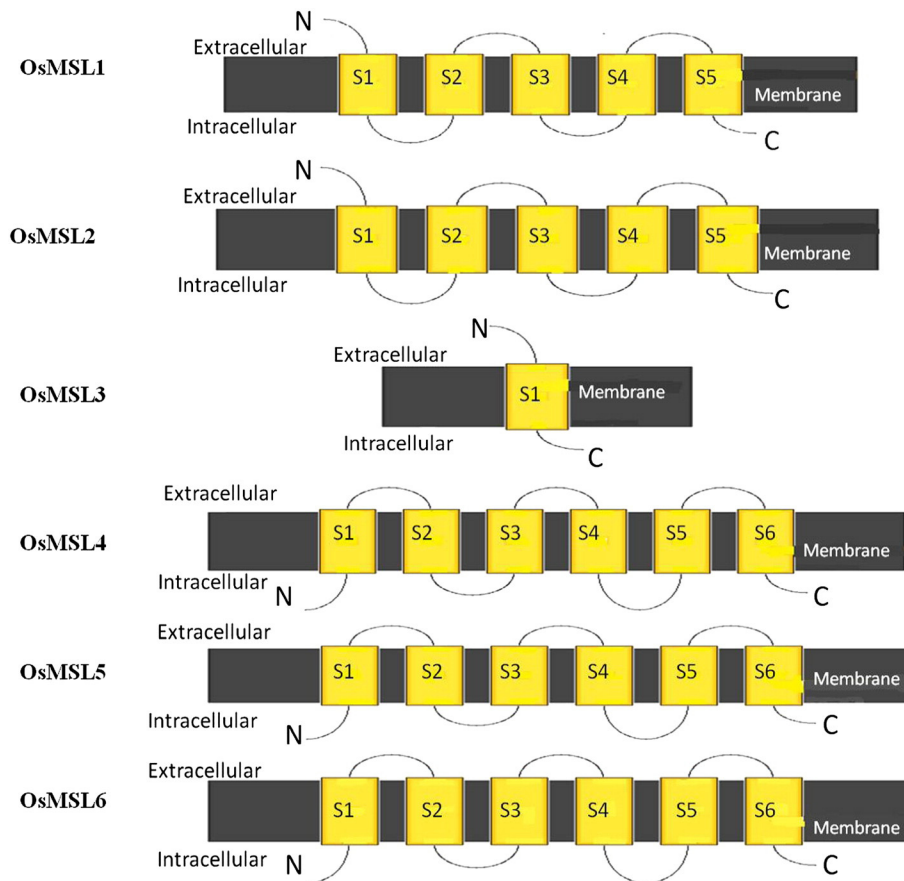


Fig. 5. A cartoon of the transmembrane helix topology summarizing the linear coordinates for the helices and indicating the protein's extra- and intercellular regions. Transmembrane helix region represented by filled yellow boxes with labels S1, S2, S3, S4, S5 and S6. Membrane regions were represented by filled black boxes. N and C terminal regions were labeled and represented as single line.

with *E. coli* MscS and its C-terminal TM domains resemble TM3 of MscS. Similar motifs have been reported in bacterial MscS channel as well. The conserved motif and consensus sequence (PF(X₁₂₋₁₆)GXV(X₂₀₋₂₁)PN(X₉)N) were recognized at the C-terminal TM domain and their surrounding sequences in *E. coli* (Pivetti et al., 2003). The MscS family has a good conservation throughout the TM3 helix which is rich in glycine and alanine residues (Bass et al., 2002; Pivetti et al., 2003; Miller et al., 2003). InterProScan analysis has combined different protein signature recognition methods into one resource including Superfamily, Pfam, Gene3D, Panther, SMART, Prosite and PIR (Protein Information Resources). The superfamily database of structural and functional protein annotations showed that the MS channel and like Sm (LSM) domain were present in all rice OsMSL sequences. The Pfam analysis (collection of protein families) confirmed the presence of MS channel domain in all OsMSL. Along with the MS channel domain, OsMSL3 has a unique ammonia transferase like plant mobile domain, transposase MuDR plant domain, zinc finger, PMZ-type domain and zinc finger SWIM-type domain. Protein information resource (PIR) database showed that rice OsMSL4, OsMSL5 and OsMSL6 have a unique MscS-like plant/fungi domain which is restricted to the plant and fungi lineages. To evaluate the signature motif of all OsMSL, we performed a MEME suit analysis and represented the distribution of motif to respective OsMSL gene (Supplementary information Fig. S1). Motif 6 is the most commonly occurring sequence and is present in all OsMSL genes. Motif 1 consists of the MS channel domain, uniformly distributed to all OsMSL except OsMSL3. Motif 10 is restricted to OsMSL1 and OMSL2. Motifs 2, 3, 5, 7 and 9 were present in OsMSL4, OsMSL5 and OsMSL6. Motifs 4 and 8 were distributed

to OsMSL3, OsMSL4, OsMSL5, and OsMSL6 (Fig. 7a). The motif sequences and residue counts were diagrammatically represented in Fig. 7b. Motifs 2 and 3 showed a protein kinase C phosphorylation site. As predicted, OsMSL4, OsML5 and OsMSL6 were located in plasma membrane, which might be regulated by phosphorylation by protein kinase C.

3.4. Gene expression analysis

Organ-specific expression of the 6 OsMSL genes under normal growth conditions in the field were obtained from the RiceXpro database (<http://ricexpro.dna.affrc.go.jp/>). The data obtained as OsMSL expression level in different tissues and abiotic stress conditions (cold, drought and salt) of rice plant was retrieved as heat maps (Fig. 8a and b). All the OsMSL genes were expressed in various tissues/organs throughout the plant life (Fig. 8a). *In silico* gene expression analysis revealed that OsMSL3 has a uniform expression pattern in almost all tissues and organs of rice. Moreover, significant expression level of OsMSL (OsMSL1, OsMSL2, OsMSL3, OsMSL5 and OsMSL6) was observed in reproductive organs such as inflorescence, anther, pistil, ovary, embryo and endosperm, while OsMSL4 showed no significant transcript expressions in tissue specific libraries. *E. coli* RpoS sigma factor controls MscS gene expression at the transcriptional level, and activated by high osmolarity in stationary phase (Stokes et al., 2003). The MS channel gene family protects prokaryote from hypoosmotic stress and act as safety valve (Martinac et al., 1987; Booth et al., 2005). Eukaryotic MSL proved to have a diverse distribution and function besides serving as a

Class I

TM

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OsMSL3      AISLFMELLGFNTQKWITAGGFGTVLLTLAGREIFTNFLSSVMINATRPFVNEWINTKI 420
AtMSL2      AVSLFMELLGFSTQKWLTAGGLGTVLLTLAGREILTNFLSSVMIHATRPFVLNEWIQTKI
AtMSL3      AASLFMELLGFSTQKWLTAGGLGTVLLTLAGREILTNFLSSIMIHATRPFVLNEWIQTKI
MscS        TLIAALGRVGVQTASVIAVLGAAGLAVGLALQGSLSNLAAGVLLVMFRPFAGEYVDL--
AtMSL1      GLMASAEACGVAVQSILTVGGVGVATAFAARDILGNVLSGLSMQFSRPFSSMGDTIKA--
OsMSL1      GVMALAEACGVAAQSILTVGGVGVATAFAARDVLGNMLSGFSLQFSSPFKAGEYIKA--
OsMSL2      GGMALAEACGVVPVQSILTVGGVGVATAFAARDVLGNILSGLSLQFSKPFVLVGDNIKA--
               * . . . : : * . : : * : : * . . . : ** : : .
               PF-X (12-16) --

OsMSL3      DGVEVSGIVEHVGWWSPTIIRGDDREAIYIPNHKFTVSILRNNTQRTHWRIK-TYLALSH 480
AtMSL2      EGYEVSGTVEHVGWWSPTIIRGEDREAIHIPNHKFTVNVRNLTQKTHWRIK-THLAISH
AtMSL3      GGYEVSGTVEHVGWWSPTIIRGDDREAVHIPNHQFSVNIVRNLTQKTHWRIK-THLAISH
MscS        --GGVAGTVLSVQIFSTTMR-TADGKIIVI PNGKIIAGNIINFSRE-PVRRNEFIIGVAY
AtMSL1      --GSVEGQVIEMLTSTLL-NAEKFPVLV PNSLFSSQVIVNKSRA-QWRAIASKIPLQI
OsMSL1      --GSIEGKVIEIGLTSTELM-NPEQLPVTVPNSLFSSQVIVNRSRA-KWRSNVTKIPIRI
OsMSL2      --GSIEGKVIEIGLTSTLLI-NPENLPVVVPNSLFSSQVIVNKSRA-VWRARVVKIPVII
               : * * : : : : : : : : : : : : : : * : :
               --GXV-----X (20-21)-----PN---X (9)---N

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Class II

TM

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841
OsMSL4      IIVFALWLLILGIATTHFFVFLSSQVLVAVFVFGNTLKTIFEAIVFLFVMHPYDVGDRCE 900
OsMSL5      VIVIALWLSILGIATSRFFVFVSSQLLVAVFMFGNTLKTIFEAIVFLFVMHPDVGDRCE
AtMSL7      IVIVIVWLILLEIATSKYLLFLTSSQVLLAFMFGNSLKTVFESIIIFLFIHHPYDVGDRLL
AtMSL8      IVIVIVWLVLLEIASSKVLFFVSSQVLLAFIFGNTVKTVFESIIIFLFIHHPYDVGDRCE
AtMSL6      I I I I I W L L I L G I T S T K F L V M S S Q V V V A F I F G N M C K I V F E S I I Y L F V I H P D V G D R C E
AtMSL4      I V I I W L L I L G I A T T K F L L V I S S Q L L V V F V G N S C K T I F E A V I F V F V M H P D V G D R C E
AtMSL5      I I I I I W L L I L G I A T T R F L L V L S S Q L L V A F V F G N S C K T I F E A I I F L F V M H P D V G D R C E
OsMSL6      I I I I I T L L L M G I A T T K I L L V I S S Q L L V V F I F G N A C K T V F E A L I F V F I M H P D V G D R C V
AtMSL9      V I T F I V W M V L L D I A S T K L L L V F S S Q F L G L A F M I G S T C K N I F E S F M F V F V M H P D V G D R C V
AtMSL10     V V T V I W L L L L E V A T T K V L L F F S T Q L V A L A F I I G S T C K N L F E S I V F V F V M H P D V G D R C V
               : : . . : : : : : : : : : : * : . * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * :
               F---P---GD--

OsMSL4      IEDCQVVVEEMNIMTTVFLRYDNLKIYYPN SVLATKPI MNYYRSPDMGEGIDFSVHVATP 960
OsMSL5      VDGMQVVVEEMNIMTTI FLRYDNLKVYYPN SQLAIQPI MNYYRSPDMGDAVDFSVHVATP
AtMSL7      IDTVEMVVEEMNILTTFVFLRADNLKIVYPN ILLWQKAIHNYRSPDMGDEVTCVHIITP
AtMSL8      IDSQVLVVEEMNILTTFVFLRYDNLKIMYPN SLLWQKAIHNYRSPDMGDAIEFCVHIITP
AtMSL6      IDGVQMVEEMNILTTFVFLRFDNQKVVPN SLLWTKSIGNYRSPDMGDGIEFSIHITTP
AtMSL4      IDGVQMIVEEMNILTTFVFLRFDNQKIVYPN SLLGTPKPI ANYYRSPDMQDAIEFVHIATP
AtMSL5      IDGVQLVVEEMNILTTFVFLRYDNQKI IYPN SVLGTPKPI ANYYRSPDMGDAIEFCVHIATP
OsMSL6      IDGIQMVVEEMNILTTFI FLKNDNEKVYYPN SVLSTKAI SNFYRSPNMYDTINFTIDVSTS
AtMSL9      VDGVMVLVEEIDLLTTVFLKIDNEKVFPN SVLISKPI SNFYRSPDMGDYVDFGIAFSTP
AtMSL10     VDGVAMLVEEMNLLTTVFLKLNNEKVYPN AVLATKPI SNFYRSPNMGETVEFSISFSTP
               : : : : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * : * :
               X (10-14) V-----X (20-21)-----PN---X (7) - IXNXXR

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Fig. 6. Alignment of class I and class II OsMSL gene family in rice. Green color highlighted part shows transmembrane region and red color highlight exhibit conserved motif. MscS is mechanosensitive channel small of *E. coli*.

safety valve. The *Arabidopsis* MSL genes were expressed in all major tissues throughout their life (Haswell, 2007). AtMSL7 was detectable at low levels only in the flower and AtMSL9 responded to mechanical and gravity stimuli (Haswell, 2007; Kimbrough et al., 2004). Stretch activated ion channel was reported in pollen protoplast and necessary for pollen tube growth (Dutta and Robinson, 2004). Expression of OsMSL genes at reproductive stages probably indicates their involvement in reproductive processes, including maturation and growth of gametes. Further extensive study will be required to shed more light on OsMSL function during reproductive stages.

To gain insight into the comprehensive roles of the OsMSL gene family members in response to various stresses, their expression patterns were investigated in rice seedlings subjected to abiotic (salt, drought and cold) stress using the Rice Oligonucleotide Array Database (ROAD) (<http://www.ricearray.org/>). In the database, 7 day-old rice seedlings were subjected to different abiotic stresses such as salt (200 mM NaCl), drought (dried at 28°C) and cold (chilled at 4°C for 3 h). *In silico* expression analysis of OsMSL demonstrated little differential expression patterns in drought, cold and salt stress as compared with the control, except OsMSL6 which exhibit significant

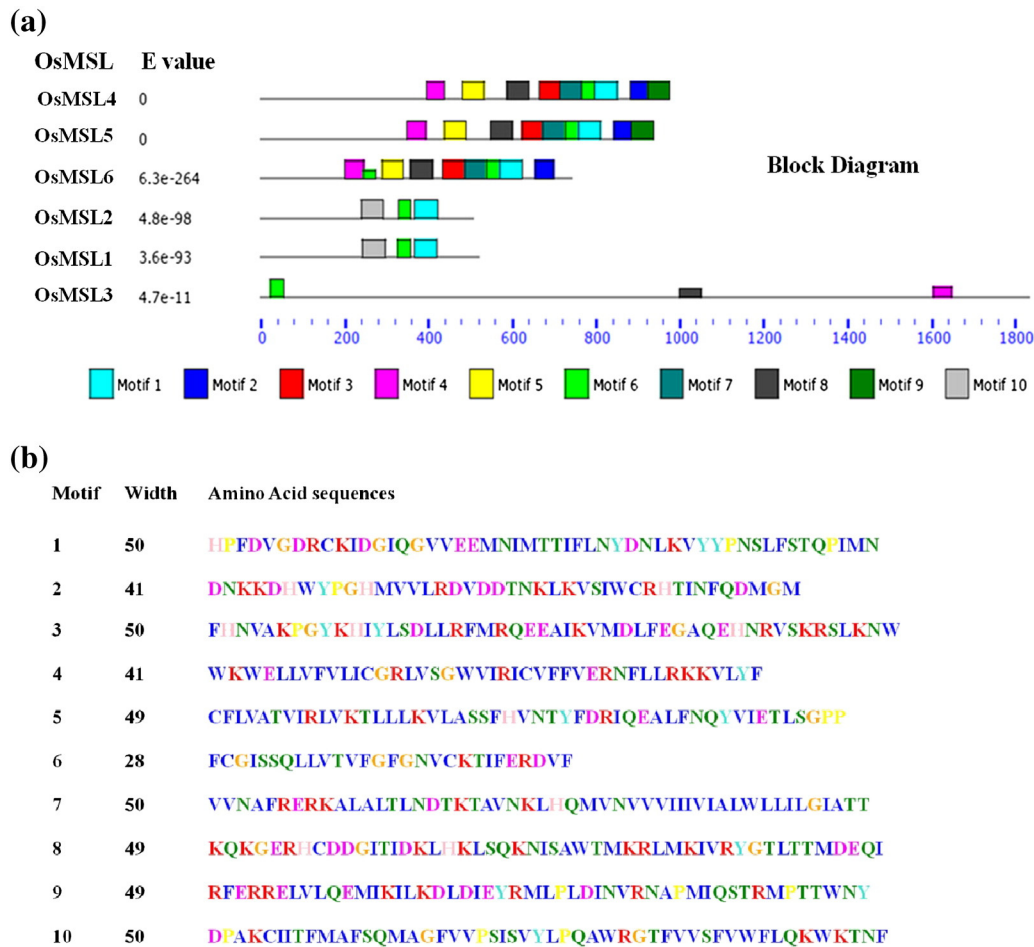


Fig. 7. Domain organization of MSL gene family from rice. (a) MEME suit analysis for the prediction of one or more motifs in OsMSL protein represented by colored block diagram. The sequences used for analysis are (OsMSL1—Os04g48940, OsMSL2—Os02g45690, OsMSL3—Os03g31839, OsMSL4—Os02g44770, OsMSL5—Os04g47320, OsMSL6—Os06g10410) (b) Representation of motif width and amino acid sequences of rice OsMSL.

expression level in drought stress. OsMSL3 exhibited significant expression pattern in both salt and drought stress (Fig. 8b). Besides these abiotic stresses, the plant recognizes and responds to diverse stimuli such as wind, submergence, temperature and turgor pressure (Knight et al., 1992). Prokaryotic cells and endosymbiotic organelles are frequently exposed to abiotic stress like hypoosmotic condition which leads to deleterious effects under normal activity (Berrier et al., 1996). Haswell (2007) predicted six members of the OsMSL gene family in rice, categorized MSL into two groups and predicted subcellular localization of the MSL in the mitochondria, chloroplast membrane or plasma membrane. Similarly in this present report, we analyzed OsMSL structure, phylogenetic relationship, localization and expression pattern. We predicted five OsMSL genes in the rice, categorized into two classes and subcellular localization of these genes were predicted in the mitochondria, chloroplast or plasma membrane. Further in this report, we predicted gene structure and transmembrane domain of the OsMSL gene family. Additionally we also analyzed *in silico* tissue specific and abiotic stress libraries for responsive OsMSL genes.

4. Conclusion

From the present study, we can conclude that the rice MSL family contains 5 genes. However, OsMSL3 is not consistent with MSL family member, indicated only a partial sequence, fusion event, or a transposon. Phylogenetic analysis classified OsMSL into two main groups, Class I (OsMSL1, OsMSL2) and Class II (OsMSL 4, OsMSL5 and OsMSL6). An *in silico* subcellular localization study concluded that

class I OsMSLs were predominantly localized either in the plastid envelope or mitochondria and class II OsMSLs were localized in the plasma membrane. Conserved motif and domains in the deduced amino acid sequence of rice MSL strongly supported their identity as members of MscS-like gene family. The topological study of OsMSL showed that, Class I and Class II OsMSL have 5 and 6 transmembrane regions respectively. The expression analysis showed the differential tissue specific expression pattern of OsMSL. The expression analysis revealed that most of OsMSL genes are expressed in the reproductive stages of rice and no significant regulation of expression level were observed during abiotic stresses. The results presented here is the first detailed study to understand the role of MscS-like genes in rice using an *in silico* approach. The information generated will be significant for further investigation of functional role of MSL genes particularly in monocot plants.

Conflict of interest

The authors declare that there is no conflict of interest.

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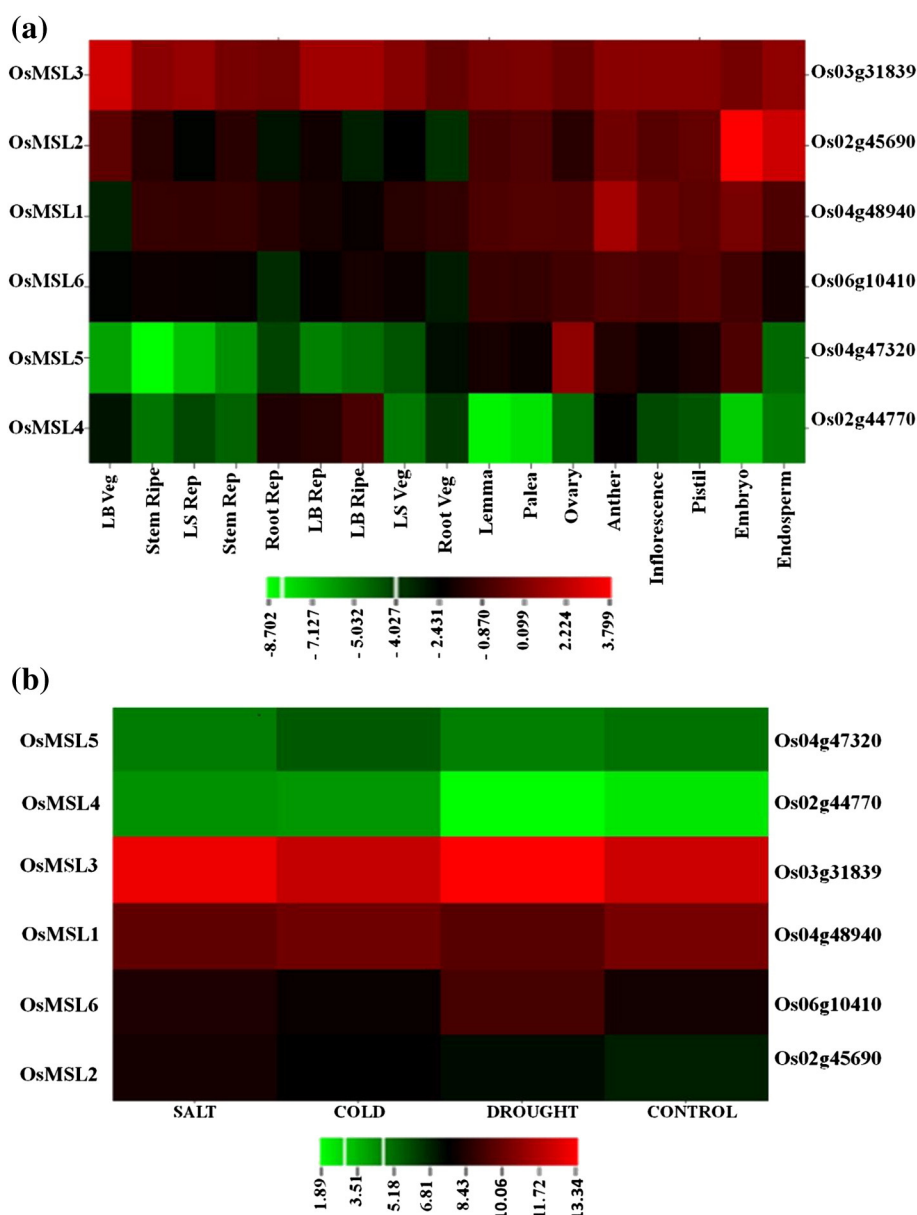


Fig. 8. Differential expression patterns of rice OMSL genes. Heat map showing differential expression profile of rice OMSL gene in (a) Tissue specific libraries. The abbreviation used in figure (LBveg—leaf blade vegetative, Stem ripe—stem ripening, LSRep—leaf sheet reproductive, Stem rep—stem reproductive, Root Rep—root reproductive, LB Rep—leaf blade reproductive, LB Ripening—leaf blade ripening, LSveg—leaf sheet vegetative, Rootveg—root vegetative). (b) Abiotic stressed (cold, salt and drought) rice tissues.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.plgene.2014.12.001>.

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